Oluwatosin Ayeni

Research Engineer, Model Inference & Serving — H Company

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PROFILE

Research Engineer specialising in end-to-end LLM inference stack ownership and multimodal serving. Benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL — 1,443× TTFT speedup via RadixAttention, 5,965 tok/s peak. Built lab-vllm-omni covering all 5 output modalities; inference-optimizer production gateway (3.87× throughput, 61.4% cost reduction). Excited by frontier multimodal agentic inference challenges. Happy to be in London office 3 days/week; open to relocation.

CORE COMPETENCIES

Inference Engines	vLLM, SGLang, TensorRT-LLM, ONNX Runtime — benchmarking, profiling, production deployment
Multimodal Serving	vLLM-Omni, all 5 modalities (text, audio, image, video, structured output); memory pressure and scheduling characterisation
GPU & CUDA	NVIDIA Nsight Systems/Compute, CUDA kernel profiling, memory bandwidth analysis, KV cache ops, tensor parallelism
Inference Techniques	RadixAttention/prefix caching, continuous batching, speculative decoding, AWQ/GPTQ, flash attention, KV-cache-aware routing
Infrastructure	Kubernetes (HPA, GPU pools), Docker, Prometheus, Grafana, FastAPI, Python — production inference platforms

PROFESSIONAL EXPERIENCE

Jan 2025–present **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

lab-vllm-omni vLLM-Omni phased inference lab: all 5 output modalities — text, audio, image, video, structured output. Characterised throughput, memory pressure, and scheduling behaviour across modalities. GitHub

llm-inference-benchmark (H100) vLLM v0.30 vs SGLang v0.5.20, 2× H100 NVL 94 GB, Qwen3-8B BF16, TP=2. RadixAttention: 1,443× TTFT speedup (39 ms v 55,565 ms). Peak: 5,965 tok/s. SLO 100% at concurrency 128. Live results

inference-optimizer Production Inference Gateway: 3.87× throughput, 61.4% cost reduction, 64.3% cheaper multi-provider routing. GitHub

llm-inference-platform Production LLM serving: vLLM + SGLang + LiteLLM gateway on RunPod/Vast.ai, KEDA autoscaling, Prometheus/Grafana/Loki. GitHub

open-model-engine GPU-first inference benchmarking: CUDA profiling, quantisation, speculative decoding. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale